P2P Process Implementation in SAP MM for BuildRight Construction Pvt. Ltd.

This project presents an industry-grade Procure-to-Pay solution modeled on SAP MM business processes for BuildRight Construction Pvt. Ltd. The application covers the full purchasing cycle from internal material requirement to vendor settlement, supported by a Python backend, SQLite persistence, SAP-style user interface, and business control logic.

Problem Statement

Manual procurement in a construction materials business creates operational and financial risk. Demand requests may be duplicated, supplier quotations may not be compared systematically, stock receipts can be delayed in recording, and finance teams may process invoices before confirming actual goods receipt. These conditions lead to over-ordering, invoice mismatch, duplicate payment risk, and weak audit traceability. A controlled P2P process resolves these issues through structured documents, posting discipline, and integrated accounting impact.

Solution and Features

- Purchase Requisition creation for internal demand capture.
- RFQ and quotation recording for supplier enquiry and price comparison.
- Purchase Order creation with vendor, quantity, and negotiated price.
- Goods Receipt posting with inventory recognition and GR/IR accounting.
- Invoice Verification with quantity and value validation against PO and GR.
- Vendor Payment processing with liability clearance and bank posting.

Company Code	BR01	Plant	BR01 (Delhi)
Purchasing Organization	BR10	Purchasing Group	BR1
Vendor	SteelCo Supplies Ltd.	Material	STL001 - Structural Steel Rods

Tech Stack

• Frontend: HTML5, CSS3, JavaScript

• Backend: Python 3.11 web server with REST-style endpoints

• Database: SQLite for persistent transaction and posting storage

• Business Domain: SAP MM Procure-to-Pay with MM-FI style integration

• Deployment: Localhost application packaged for ZIP and GitHub publishing

Unique Points

• Implements the complete P2P lifecycle in a working full-stack application rather than static mock screens only.

• Uses realistic document sequencing for PR, RFQ, quotation, PO, goods receipt, invoice, payment, and FI documents.

• Enforces 3-way match visibility between Purchase Order, Goods Receipt, and Invoice quantity or value.

• Blocks payment when invoice variance exceeds PO or GR values, demonstrating compliance control.

• Provides SAP-style transaction workspace, document history, and accounting ledger for evaluator demonstration.

Business Value

The project improves procurement visibility, supports auditability, reduces duplicate payment risk, and demonstrates how purchasing and finance activities remain connected through accounting entries. It also creates a stronger academic submission because it combines functional software, business logic, screenshots, and documentation in one deliverable.

Application Screenshots

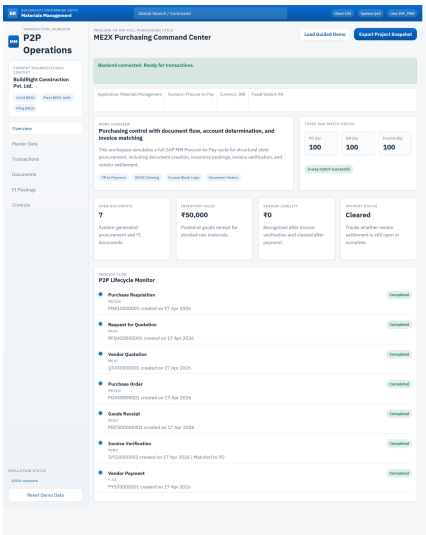

Figure 1. SAP-style overview cockpit showing process metrics and lifecycle monitor.

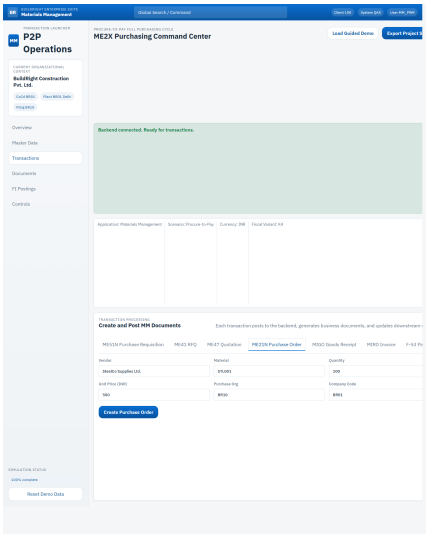

Figure 2. Transaction processing workspace used to create and post MM documents.

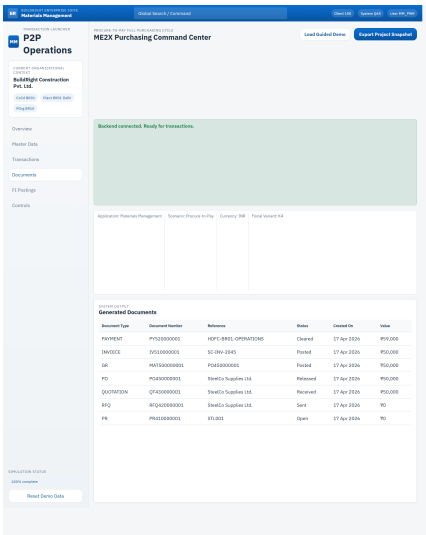

Figure 3. Generated document register with business status and reference values.

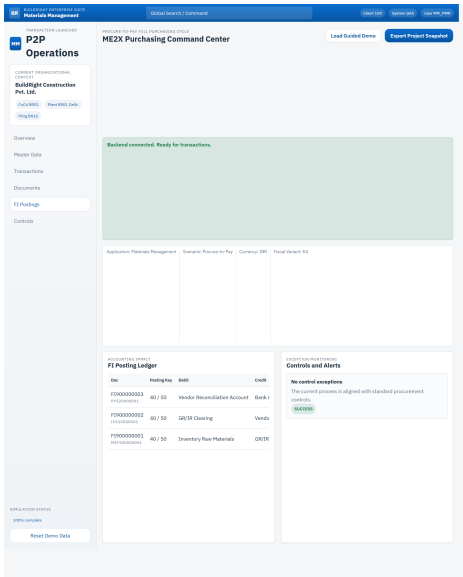

Figure 4. FI posting ledger showing inventory, GR or IR, vendor, and bank impact.

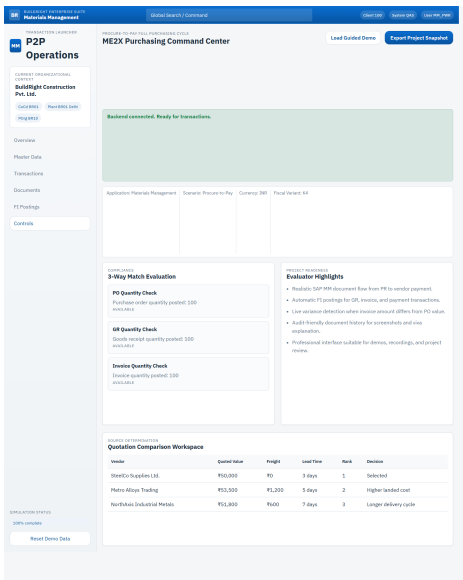

Figure 5. 3-way match controls, exception alerts, and quotation comparison.

Future Improvements

• Add user authentication, approval workflow, and role-based process control.

• Introduce searchable document drill-down pages similar to SAP list reports.

• Extend vendor evaluation with scoring for price, lead time, quality, and service.

• Add charts for spend analytics, blocked invoices, and open liabilities.

• Package the solution with deployment automation for easier public hosting and GitHub presentation.

Conclusion

The BuildRight P2P Command Center demonstrates how SAP MM procurement concepts can be translated into a functional software project with real business flow, persistent backend logic, and evaluator-friendly outputs. The final solution covers PR, RFQ, quotation, PO, GR, MIRO, and vendor payment while highlighting 3-way match control and MM-FI

integration, making it suitable for project submission, demo, and viva discussion.